A Synthetic Record-Linkage Benchmark for BOAMP Procurement-Renewal Detection

Calibration, data-generating process, validation, and current readiness

Benchmark version `v0_3_temporal_candidate_revision`, central scenario `central_provisional`

BOAMP renewal-linkage and survival-analysis study

Generated 2026-07-30 from the artifacts listed in Section 23

Abstract

Real BOAMP notices carry no field stating that one contract renews another, so no direct measurement of linkage precision or recall is possible on them. This report documents a synthetic benchmark built to answer the narrower question that *is* answerable: whether a candidate linkage method recovers *known* recurrence relations in a population whose observable structure has been calibrated to BOAMP. A latent procurement world — buyers, establishments, recurring needs, contract cycles, and the successor relation between them — is generated first; BOAMP-like publication notices are then emitted from it and subjected to identifier, name, CPV, duration, and text corruption whose rates are estimated from the real corpus conditionally, not marginally. A sealed truth layer is retained and never exposed to any linkage code.

The current central world contains 87,360 observed notices drawn from 19,200 latent buyers and 63,291 contract cycles, with 291,855 logged corruption events. Across 231 metrics in 15 gates the validation framework reports `PASS_WITH_WARNINGS`, with 5 metric-level failures, all in non-critical gates. Every generated artifact (51/51) reproduces bit-for-bit from its recorded seeds and configuration.

The separate five-level readiness assessment is the operative statement of what the benchmark may be used for, and it is deliberately conservative: preliminary synthetic-only work is supported with limitations, while controlled algorithm comparison, final algorithm ranking, and release as a validated benchmark all currently fail. The binding reasons are seed-to-seed instability of the difficulty metrics and unstable probe rankings, not defects in the generator's internal consistency. Nothing in this report estimates real BOAMP recurrence prevalence or real-world linkage accuracy, and the benchmark is not a substitute for the real ground truth that does not exist.

Contents

1 Executive summary

What this is. A controlled, fully reproducible synthetic corpus of French public-procurement notices in which the answer to "does notice j continue the contract that notice i announced?" is known by construction, and in which the observable record — missing identifiers, drifting buyer names, absent CPV codes, boilerplate text — has been made to resemble the real BOAMP corpus of 84,623 notices published between 2015-03-02 and 2026-07-13.

Why it exists. The production pipeline links a source contract to a later notice when a weighted similarity score clears a frozen threshold. On real BOAMP that produces 1,003 accepted links from 3,380 eligible sources (29.7%) — but there is no way to say how many of those are correct, because no legal-renewal field exists. Every quality number the production pipeline can report is either model-based or conditional on its own accepted links. The benchmark supplies the missing denominator, in a world whose realism is a documented, measured claim rather than an assumption.

What the current run establishes.

- **Internal correctness.** All 8 critical gates that concern internal consistency, specification recovery, candidate environment, privacy, and algorithm utility pass. No truth column, alias, normalised truth value, or hash of a truth identifier reaches the observed table.
- **Observable realism, conditionally.** The candidate environment — the property that decides whether a linker can find the successor at all — now matches the real corpus closely: zero-candidate rate 42.4% synthetic against 40.7% real, a gap of 1.69 percentage points.
- **Difficulty is non-trivial and measured.** Of 1,807 true successor pairs inside the algorithm's own scope, the production 6-month blocking window reaches 0.392; widening to 36 months reaches 0.956. The best frozen probe linker attains pair $F_1 = 0.316$ — far from both 0 and 1, which is what makes the benchmark discriminative.
- **Reproducibility.** 51 of 51 generated artifacts replay to identical canonical content hashes.

What it does not establish. Seed-to-seed spread is material: production blocking completeness over 10 central seeds has mean 0.391 and standard deviation 0.033 (coefficient of variation 0.083), and probe rankings flip between seeds. Consequently `READY_FOR_CONTROLLED_ALGORITHM_COMPARISON`, `READY_FOR_FINAL_ALGORITHM_RANKING` and `READY_FOR_VALIDATED_SYNTHETIC_BENCHMARK_RELEASE` all read `FAIL`. The benchmark is currently an instrument for debugging and integrating linkage code, not for ranking linkage algorithms.

2 Research objective and benchmark purpose

The parent study models time to renewal of French public digital/ICT procurement contracts. Its event is not observed: BOAMP publishes notices, not contract lineages. The study therefore *constructs* the event by record linkage — for each eligible source contract, later notices from the same buyer near the estimated contract end are scored, and the best-scoring candidate is accepted if it clears a threshold. Every survival conclusion inherits whatever error that construction makes.

That error is unmeasurable on real data. The purpose of this benchmark is to make it measurable in a controlled setting, and to state precisely how far a measurement made there transfers. Formally, write $\mathcal{L}$ for a linkage procedure, $\mathcal{R}$ for the real BOAMP corpus, $\mathcal{S}$ for a synthetic corpus, and T for the true successor relation. On $\mathcal{R}$, T is unobserved and no estimate of $\Pr(\hat{T} = T)$ exists. On $\mathcal{S}$,

T is known by construction, so precision, recall, and cluster-level accuracy are computable exactly. The transfer claim is conditional and is the only one this report makes:

> *If $\mathcal{S}$ reproduces the observable properties of $\mathcal{R}$ that materially affect $\mathcal{L}$, then the behaviour of $\mathcal{L}$ on $\mathcal{S}$, evaluated across a documented envelope of scenario assumptions, is useful calibration and comparison evidence about its behaviour on $\mathcal{R}$.*

The design follows the gold-standard-and-corruption framework of Lam et al. [9], adapted from personal identifiers to procurement recurrence: generate a clean population with known links, then corrupt the observable layer with mechanisms whose rates are calibrated to the target corpus, and evaluate against the retained truth. The linkage vocabulary — match, potential match, non-match, blocking, pairs completeness, reduction ratio — follows Fellegi and Sunter [4] and Christen [2], and the design of evaluating linkage software on constructed data follows Ferrante and Boyd [5].

3 Scope and limitations

In scope. A single geographic and sectoral slice: Pays de la Loire buyers, digital/ICT contracts, publication dates 2015-03-02 to 2026-07-13. One central scenario plus bracketing scenarios. Notice types are reduced to calls (`APPEL_OFFRE`) and awards (`ATTRIBUTION`); the residual `OTHER` category of the real corpus (4.5% of notices) is not generated.

Out of scope, and never claimed.

1. **Real recurrence prevalence.** The generator's recurrence propensity is an assumption, not an estimate. See Section 7.
2. **Real linkage accuracy.** Every precision and recall number in this report is measured against generated truth.
3. **Threshold optimality.** The production threshold is an output of a particular pipeline; it is reported for comparison and never used to define truth.
4. **National generalisation.** The calibration corpus is one region.
5. **Held-out validation.** No real-BOAMP partition was withheld from generator design, so observable-fidelity results are in-sample with respect to calibration.

4 The original BOAMP data

Source and extraction. Monthly JSON extracts from the *Bulletin officiel des annonces des marchés publics* were retrieved via the Opendatasoft Explore API, filtered server-side to Pays de la Loire, for publication months 2015-03-02 through 2026-07-13. After flattening and de-duplication on the BOAMP notice identifier, the prepared corpus contains 84,623 unique notices with publication dates from 2015-03-02 to 2026-07-13.

Schemas. Two incompatible record schemas coexist. The legacy BOAMP format accounts for 87.4% of notices; the EU eForms format, mandated from late 2023 and adopted gradually from 2024, accounts for 12.6%. The two differ in which fields exist at all, which is why every missingness rate in this report is reported conditionally on schema and never only marginally.

Notice types. Normalised notice type splits 68.9% calls, 26.7% awards, and 4.5% other (modifications, cancellations, corrections).

Analysis population. Applying the digital/ICT scope — CPV divisions 32, 35, 48, 72, or a French-language keyword match on the cleaned contract object — to call notices yields 3,380 eligible source contracts, the population every production and benchmark comparison uses.

5 Analysis population and preprocessing

Preprocessing is layer-neutral and runs once (`notebooks/01_data_engineering_pipeline.ipynb`, Part C). It performs, in order: de-duplication on the notice identifier; notice-type normalisation; schema-family detection; CPV extraction across both schemas, including the UBL commodity-classification paths eForms uses; buyer-identifier extraction and validation; buyer-name normalisation; text cleaning; and duration extraction with a plausibility filter of 1–120 months.

Two derived quantities carry disproportionate weight and are flagged here rather than buried.

Buyer key. Buyer identity is resolved by a strict priority ladder: a checksum-valid SIRET, else a checksum-valid SIREN, else the normalised buyer name. Only 27.1% of notices carry a checksum-valid SIRET, so the large majority of buyer identities rest on a normalised string. This is the single largest source of both false splits and false merges, and the benchmark models both explicitly.

Estimated contract end. Only 13.7% of notices declare a duration at all. Missing durations on eligible sources are imputed by the median observed duration of the same CPV division. The imputed duration then determines the estimated end date, which determines the temporal blocking window, the temporal score component, and the survival covariate — a dependency chain documented as the pipeline's highest-leverage assumption.

6 Calibration methodology

Principle. A quantity may be used to *set* a generator parameter only if it is measurable from raw or layer-neutral prepared fields, without reference to any accepted link, score, or threshold. Quantities visible only through accepted links are silver-standard: they may be reported and used as validation targets, never as generator inputs. Quantities not measurable at all become scenario assumptions and must be varied.

Estimator conventions. Throughout, $\widehat{p}$ denotes an empirical proportion computed on a stated population, and p a model probability the generator draws from. For a binary outcome Y observed on cells c defined by conditioning variables, the generator consumes a beta-binomially smoothed rate rather than the raw cell proportion:

$$\widehat{p}_c = \frac{\sum_{i \in c} Y_i + \kappa \widehat{p}_\bullet}{n_c + \kappa}, \qquad \widehat{p}_\bullet = \frac{1}{n} \sum_{i=1}^{n} Y_i, \tag{1}$$

with prior strength $\kappa = 25$ notices and $\widehat{p}_\bullet$ the global rate. Shrinkage matters because the conditioning grid (schema $\times$ year $\times$ notice type) contains cells with a handful of notices, whose raw proportions are 0 or 1 and would otherwise be copied into the generator as certainties.

Distances, not tests. Agreement between a real and a synthetic distribution is judged by effect size, never by a p-value. On 84,623 real notices any non-zero difference is statistically significant, so significance carries no information about whether the benchmark is fit for purpose. For a categorical variable with real distribution P and synthetic distribution Q over categories $\mathcal{K}$ the report uses total variation and Jensen–Shannon distance [10]:

$$\mathrm{TV}(P,Q) = \tfrac{1}{2}\sum_{k\in\mathcal{K}} |P_k - Q_k|, \qquad \mathrm{JS}(P,Q) = \tfrac{1}{2}\mathrm{KL}(P \,\|\, M) + \tfrac{1}{2}\mathrm{KL}(Q \,\|\, M)\,, \quad M = \tfrac{1}{2}(P+Q). \tag{2}$$

Both are bounded in $[0,1]$, require no distributional assumption, and are interpretable as "the largest probability mass the two distributions disagree about" and "an information-theoretic divergence, symmetrised and bounded" respectively. For continuous variables the report uses a scale-free first Wasserstein distance, W_1 divided by the real interquartile range, so a duration discrepancy and a text-length discrepancy are comparable. For a conditional rate the report uses the count-weighted mean absolute error over cells,

$$\mathrm{WMAE} \;=\; \frac{\sum_c n_c \left|\widehat{p}_c^{\,\mathrm{syn}} - \widehat{p}_c^{\,\mathrm{real}}\right|}{\sum_c n_c}, \tag{3}$$

expressed in percentage points, which prevents a near-empty cell from dominating the verdict.

Equivalence, not non-rejection. Where a cluster bootstrap is available (200 replicates, resampling buyers rather than notices so within-buyer dependence is respected), a metric is declared equivalent only if the whole bootstrap interval lies inside the tolerance band, following the two-one-sided-tests logic of Lakens [7], Lakens et al. [8]. Metrics without an interval can reach at most "not obviously discrepant". This distinction is preserved in the metric table and is why a gate can read `INCONCLUSIVE` rather than `PASS`.

7 Parameter-provenance framework

Every generator parameter and every validation rule belongs to exactly one provenance class, recorded machine-readably in `registries/parameter_registry.csv` with 603 rows. The registry records, per row: the mathematical definition, the value or distribution, the scenario-file value, the effective value actually used, the estimation population, the estimating code, the uncertainty, whether the row may be used for calibration, whether it must remain held out, and its known limitations.

Table 1: Parameter-provenance classes and how many registry rows carry each. Source: `registries/parameter_registry.csv`.

Provenance class	Meaning	Rows
`EMPIRICAL_OBSERVABLE`	estimated from a measurable real-BOAMP quantity	273
`SILVER_STANDARD_APPROXIMATION`	estimated through a documented imperfect proxy	28
`SCENARIO_UNIDENTIFIED`	assumption about a quantity BOAMP cannot identify	206
`FIDELITY_TARGET`	observable property used only to judge realism	5
`ALGORITHM_PARAMETER`	linkage or blocking setting, never synthetic truth	77
`IMPLEMENTATION_CONSTANT`	technical choice with no empirical interpretation	14

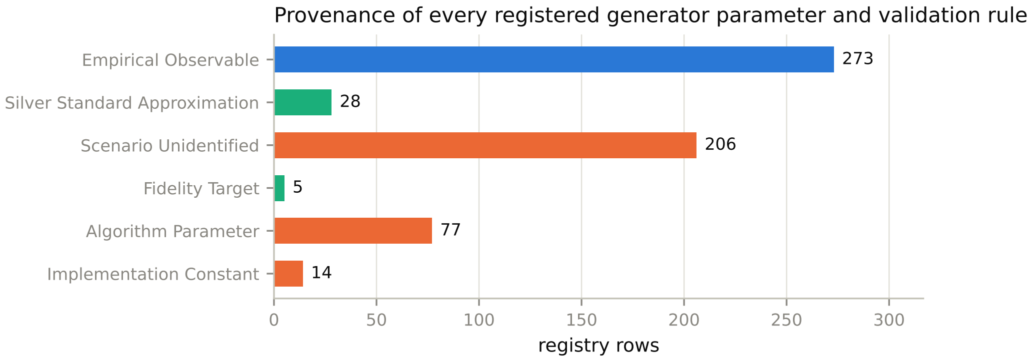

Figure 1: Provenance of every registered parameter and validation rule. The point of the figure is the ratio: a substantial part of the generator is *assumed*, not estimated, and those rows are exactly the ones that must be varied across scenarios rather than quoted as findings. Source: `scripts/build_report_figures.py`.

The distinction that matters most. A parameter swept until the synthetic candidate environment resembled the real one is *not* an empirical estimate of the real renewal process. The v0.3 scoped-candidate block (Section 9.2) is of this kind: its five values were selected by a bounded sweep against candidate-count fidelity gates. They are recorded as tuned-for-fidelity parameters, they change linkage difficulty, and they are reported nowhere in this document as measurements of BOAMP behaviour.

8 Statistical description of the original data

Table 2 lists the headline calibration quantities the generator consumes directly. Each is an empirical proportion on the full prepared corpus unless stated otherwise; each carries its own limitations entry in `config/synthetic/calibration_parameters_v0_1.yaml`.

Table 2: Directly consumed empirical calibration quantities. Population: all 84,623 prepared notices unless noted. "Observed" means what a reader of BOAMP sees — not a property of any latent entity.

Quantity	Value	Definition and unit
Checksum-valid SIRET present	27.1%	share of notices whose buyer SIRET passes format and Luhn validation
CPV missing	17.1%	share of notices with no extractable CPV code
Duration present	13.7%	share of notices declaring any raw duration field
Exact text duplication	51.4%	share of cleaned contract objects appearing verbatim more than once
Buyer-name variation	27.9%	share of validated SIRETs appearing under more than one normalised name (false-split risk)
Buyer-name ambiguity	7.6%	share of normalised names mapping to more than one legal buyer (false-merge risk)
Generic buyer name	36.8%	share of notices whose normalised buyer name is on the generic stoplist
eForms share	12.6%	share of notices in the eForms schema family
Award probability	0.388	awards per call, used as the per-cycle award-emission probability

Why marginals are not enough. Three of these quantities are strongly conditional, and a generator matching only the marginal would model the wrong mechanism.

- **Identifier availability** shows a regime shift centred on 2022 that is independent of the schema cutover: 2022 and 2023 are both entirely legacy-schema in this corpus, yet SIRET presence jumps from roughly one notice in ten before 2022 to more than half in 2022–2023, then partially recedes. Figure 7 shows the real curve and the generator's fit.
- **CPV missingness** is essentially a schema property: eForms mandates the field, legacy does not, and within legacy, calls omit CPV about three times as often as awards.
- **Duration presence** is near-deterministic in schema: essentially zero for every year through 2023, then substantial once eForms is adopted.

Buyer activity. Notices per buyer are extremely concentrated — a heavy tail in which a small number of buyers account for most publications. Figure 6 gives the rank-frequency and Lorenz views. This matters for linkage because candidate density is a function of buyer activity: a linker's hardest cases are concentrated in the tail.

Candidate environment. Applying the production blocker to the real corpus produces a median of 1 candidate per source and a mean of 3.21, with 40.7% of eligible sources receiving no candidate at all. That zero-candidate rate is structural censoring: those sources cannot be linked by any scoring rule, and reproducing it is a prerequisite for the benchmark to be informative about blocking.

9 Synthetic benchmark design

9.1 Two layers, and why the separation is load-bearing

The generator emits two disjoint sets of tables (Table 3). The *observed layer* is a single BOAMP-like notice table with 12 columns, containing only fields a linkage method could read. The *truth layer* carries the latent entities, the cycle chains, the successor relation, per-notice pre-corruption values, and a complete corruption log — 36 truth-only columns across 9 tables in total.

The separation is what makes exact blocking recall, recovery by corruption type, and cluster-level accuracy computable. It is also why leakage is treated as a release-blocking failure: a single truth column reaching the observed table would turn every accuracy number downstream into an artifact. The leakage check is not a column-name blocklist alone; it also rejects unsuffixed hidden-truth aliases, exact and normalised truth-identifier *values*, and MD5, SHA-1 and SHA-256 hashes of truth identifiers.

Table 3: Emitted tables, the layer each belongs to, and its statistical unit. Only the observed layer is visible to a linkage method. Source: `registries/unit_of_analysis.csv`.

Table	Layer	Grain	Rows	Cols
`observed_notices`	observed	notice	87,360	12
`clean_notices`	truth	notice	87,360	17
`notice_family_membership`	truth	notice	87,360	3
`true_relations`	truth	cycle	63,291	9
`latent_cycles`	truth	cycle	63,291	8
`latent_needs`	truth	need	44,063	9
`latent_buyers`	truth	buyer	19,200	11
`latent_establishments`	truth	establishment	20,811	7
`corruption_log`	truth	notice-field-event	291,855	8

9.2 The scoped-candidate block

Version 0.2 of the generator produced a technically valid world that was useless as a linkage benchmark: 90.4% of its digital-scope sources had no candidate at all, against 40.7% in the real corpus. The cause was not corruption but timing — true successor gaps drawn from a distribution centred well outside the production window, combined with too few same-buyer distinct needs to create distractors.

Version 0.3 adds a *scoped-candidate block* that acts only on needs whose CPV division falls in the digital scope. It has five tuned parameters, all recorded as fidelity-tuned rather than empirical: a recurrence-propensity multiplier of 1.60, a near-window share of 0.90, a hard-negative chain-alignment rate of 0.00, a scoped-buyer affinity share of 0.102, and a high scoped-need probability of 0.54. Section 18 reports what they achieved.

Two properties of this block are worth stating explicitly, because they bound what it can be accused of. It never assigns per-source candidate counts, scores, accepted links, or thresholds — it acts on latent recurrence timing and on how needs cluster within a buyer, and candidate counts remain a *post-hoc* fidelity target. And chain alignment translates an entire distinct-need chain by one constant offset, so target start − source expected end is unchanged by construction: the step cannot move the true-gap distribution at all.

10 Mathematical specification of the data-generating process

10.1 Notation

Table 4: Notation. A $*$ superscript marks a latent quantity that also has an observed counterpart; in the emitted tables the same distinction is carried by the `_true` column suffix.

Symbol	Type	Meaning
$b = 1, \ldots, B$	index	latent buyer; $B = 19{,}200$
e	index	latent establishment, nested in a buyer
d	index	latent procurement need, nested in a buyer
k	index	contract cycle within a need, $k = 1, \ldots, K_d$
ν	index	published notice
a_b	latent	buyer activity weight, $\sum_b a_b = 1$
λ_b	latent	expected number of needs for buyer b
ρ_d	latent	recurrence propensity of need d
μ_d	latent	duration profile of need d (months)
S^*_{dk}	latent	start date of cycle k of need d
D^*_{dk}	latent	true duration of cycle k (months)
E^*_{dk}	latent	expected end, $S^*_{dk} + D^*_{dk}$
G_{dk}	latent	gap from E^*_{dk} to $S^*_{d,k+1}$ (months)
T	latent	successor relation; $(dk) \to (d, k{+}1)$ when realised
Q_ν	latent	record-quality class of notice ν
$m(Q_\nu)$	latent	severity multiplier for class Q_ν
π_b	latent	identifier-quality propensity of buyer b
X_ν	observed	the emitted notice record
$\widehat{p}_c$	estimate	smoothed empirical rate in conditioning cell c
τ	algorithm	production acceptance threshold, $\tau = 0.343167$
W	algorithm	blocking half-window, $W = 6$ months
S_{ij}	algorithm	composite candidate score for source i, candidate j

Two independent pseudo-random streams are used throughout: $\omega \sim$ PCG64(world seed) for everything latent, and $\varepsilon \sim$ PCG64(corruption seed) for everything observational. Holding ω fixed while varying ε therefore re-observes the same latent world, which is what makes "the same truth under a different observation process" a well-defined experiment.

10.2 Generation order

Each step conditions only on steps above it, which is what makes the chain replayable from its seeds.

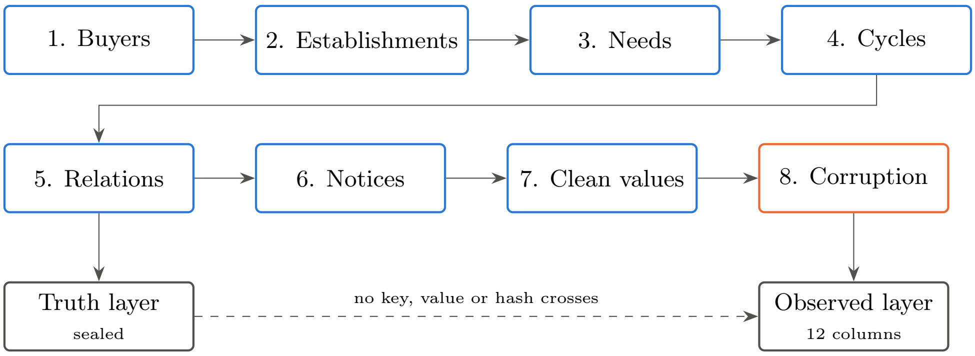

Figure 2: Generation order and layer separation. Blue stages draw from the world stream ω; the orange stage draws from the corruption stream ε. The dashed edge is the invariant the privacy and leakage gates enforce.

10.3 Step 1 — buyers

Activity weights are drawn from a Pareto variate and normalised, so concentration is emergent rather than assigned:

$$r_b \sim \text{Pareto}(\alpha) + \delta, \qquad a_b = \frac{r_b}{\sum_{b'} r_{b'}}, \qquad \alpha = 1.22,\ \delta = 0.15. \tag{4}$$

Here Pareto(α) is NumPy's parameterisation, whose support is $[0, \infty)$ with density $\alpha(1 + x)^{-(\alpha+1)}$ — the Lomax form, not the Pareto Type I form assumed by the textbook identity Gini $= 1/(2\alpha - 1)$. That identity is asymptotic and requires $\alpha > 1$; at $B = 19{,}200$ it is a poor guide, so α was set by simulating the emergent Gini rather than by inverting the formula. The offset δ keeps every weight strictly positive before normalisation and is an implementation constant.

Departments are assigned by a deficit-filling pass over buyers in descending activity, which makes the *notice-weighted* department distribution track the empirical one — a per-buyer i.i.d. draw would let one high-activity buyer distort it. Buyer type, name, alias propensity $\sim$ Beta$(2, 5)$, and identifier-quality propensity $\pi_b \sim$ Beta$(5, 2)$ complete the record. Names are syllable-generated; no real organisation name is ever copied.

10.4 Step 2 — establishments

Each buyer receives one to three establishments, more likely several for high-activity buyers. SIRENs are rejection-sampled until they pass the same Luhn validation the production pipeline applies to real identifiers; each SIRET is its buyer's SIREN concatenated with a zero-padded establishment sequence and the unique valid check digit. "Valid" therefore means the same thing on both sides of the comparison, and same-SIREN/different-SIRET ambiguity exists structurally for the corruption step to exploit.

10.5 Step 3 — needs

The number of needs per buyer is Poisson with a mean allocated continuously by activity weight rather than by activity tier:

$$N_b \sim \mathrm{Poisson}(\lambda_b), \qquad \lambda_b = f + a_b\big(\Lambda - fB\big), \qquad \Lambda = \sum_b \lambda_b^{\mathrm{tier}}, \tag{5}$$

where f is a small floor keeping low-activity buyers viable and Λ is the total scale implied by the tier means. Collapsing this to three tier means — the earlier design — cut the needs-per-buyer Gini from roughly 0.8 to 0.34 and destroyed the dense candidate neighbourhoods that make linkage hard.

Each need draws a CPV division (from the empirical division distribution, or from the scoped mixture below), a synthetic eight-digit CPV code, a division-conditioned concept and vocabulary set, a duration profile

$$\mu_d \sim \mathrm{LogNormal}\big(\log 12,\ 0.7^2\big) \ \text{ clipped to } [1, 120] \text{ months}, \tag{6}$$

and a recurrence propensity jittered multiplicatively around the scenario central value $\rho_0 = 0.40$:

$$\rho_d = \min\big(0.98,\ \rho_0 \cdot J_d\big), \qquad J_d = 0.6\,\mathrm{Beta}(4,4) + 0.7 \in [0.7, 1.3]. \tag{7}$$

No need is ever given the scenario value exactly; heterogeneity in recurrence is part of the design rather than noise added afterwards.

When the scoped block is active, the division draw becomes a two-component mixture. A share 0.102 of buyers are marked as digital-affinity buyers; their needs fall in the scoped divisions with probability 0.54, and every other buyer's needs do so with the residual probability that preserves the overall scoped mass. This clusters digital needs within buyers — which is what generates same-buyer distinct-need distractors — without changing the aggregate CPV composition.

10.6 Step 4 — cycles, and the censoring that shapes the truth

For each need, cycle 1 starts uniformly inside its buyer's active window. Each cycle draws a true duration around the need's profile,

$$D^*_{dk} \sim \mathcal{N}\big(\mu_d,\ (0.25\mu_d)^2\big) \ \text{ clipped to } [1, 120], \qquad E^*_{dk} = S^*_{dk} + 30.44 \cdot D^*_{dk} \text{ days}. \tag{8}$$

A successor is realised only if a Bernoulli draw succeeds *and* the resulting start falls inside the observation window:

$$\Pr(\text{successor at } k{+}1) = \min\big(0.98,\ \rho_d \cdot \gamma_d\big), \qquad \gamma_d = \begin{cases} 1.60 & d \text{ scoped}, \\ 1 & \text{otherwise}, \end{cases} \tag{9}$$

$$S^*_{d,k+1} = E^*_{dk} + 30.44\,G_{dk}, \qquad \text{realised only if } S^*_{d,k+1} \le \text{2026-07-13}, \qquad K_d \le 4. \tag{10}$$

The gap is a two-component mixture: with probability 0.90 for a scoped need it is drawn from a near-window component, otherwise from the scenario base component:

$$G_{dk} \sim \begin{cases} \mathcal{N}(2.0,\ 2.0^2) \text{ clipped to } [0.1, 6.0], & \text{scoped, w.p. } 0.90, \\ \mathcal{N}(18.0,\ 9.0^2) \text{ clipped below at } 1.0, & \text{otherwise.} \end{cases} \tag{11}$$

Note the implemented form: these are *clipped* normals, not renormalised truncated normals. Mass falling outside the bounds is placed on the bounds rather than redistributed, so the realised law has atoms at 1.0 and at 0.1 and 6.0. Writing $\mathcal{N}$ "truncated" in the textbook sense would misdescribe the code.

The censoring is not a detail. Because a chain stops when $E^*_{dk} + G_{dk}$ leaves the window, the gaps that *reach* the truth table are systematically shorter than the declared law of Equation (11). Any recovery check that simulates the unconditional draw will therefore fail for reasons that have nothing to do with the generator. The specification-recovery gate censors each simulated gap by the headroom remaining to its own source cycle, measured from E^*_{dk} — not from S^*_{dk}, which was an earlier error that overstated headroom by one cycle duration and biased the reference interval upward. The realised mean true gap is 15.55 months and lies inside the corrected 99% Monte Carlo interval.

10.7 Step 5 — the successor relation

The truth table is a pure function of the realised cycle chains: cycle k of need d emits `NEXT_CYCLE` to cycle k+1 if that row exists, and `NO_SUCCESSOR` otherwise, with $G_{dk} = (S^*_{d,k+1} - E^*_{dk})/30.44$ recorded in months. Nothing is hidden in this step: the three distinct reasons a chain can stop — failed Bernoulli draw, cycle cap, window boundary — are all recorded as the same `NO_SUCCESSOR`, exactly as an analyst of real data would see them.

10.8 Steps 6–7 — notices and clean values

Each cycle emits a call notice at S^*_{dk}, and with probability 0.388 an award notice after a delay $\sim \mathcal{N}(90, 45^2)$ days clipped to $[10, 400]$. The award delay is an acknowledged design assumption: no observable call-to-award distribution was carried into the calibration file.

Schema family is drawn probabilistically, not by a date cutoff: every notice before 2024 is legacy, and from that year a notice is eForms with the empirically observed adoption share for its year. An earlier hard cutoff at the mandate date gave every post-mandate notice a 100% eForms label when the real share never exceeds 60%, which then propagated into the schema-conditional duration and CPV mechanisms.

Contract text is rendered from templates over the need's concepts and vocabulary, with three length regimes so the length distribution has a tail. A call and its award share the same underlying text, subject to light publication-level variation at severity 0.03— the real corpus's within-family similarity is essentially 1, so this is deliberately small. Successor-cycle lexical drift is a separate mechanism at severity 0.50, and is a scenario parameter because no empirical anchor for true recurrence drift exists anywhere in the corpus.

10.9 Step 8 — observation and corruption

Every field's corruption severity is scaled by one latent per-notice record-quality class, so defects co-occur on the same notices rather than being sampled independently per field — the core corruption-design principle of Lam et al. [9]. The class is drawn per notice from the scenario mix (0.30 high, 0.45 medium, 0.25 low, before tilting), tilted by the buyer's identifier-quality propensity π_b:

$$\Pr(Q_\nu = \textsc{high}) \propto \phi_{\textsc{high}}\,(0.5 + \pi_b), \qquad \Pr(Q_\nu = \textsc{low}) \propto \phi_{\textsc{Low}}\,(1.5 - \pi_b). \tag{12}$$

The severity multiplier over (high, medium, low) is $m(Q) = (0.35, 1.00, 2.20)$ for identifiers, CPV, text and linked-notice visibility. Duration has its own, much flatter spread $(0.89, 1.00, 1.15)$. That is not a tuning convenience: with the shared multiplier it is structurally impossible to reach the real corpus's duration presence rate, because the high-quality class alone contributes more duration-present notices than the entire real target allows. The data are saying that duration is omitted near-uniformly across BOAMP regardless of overall record quality — an editorial convention — while identifiers, CPV and text do vary with record quality.

Field mechanisms, in the order applied:

- **Identifiers.** SIRET presence is drawn from the smoothed conditional rate $\widehat{p}_c$ of Equation (1) on cells (schema, year, notice type). When present, a small share is replaced by a sibling establishment's SIRET under the same SIREN — a valid identifier pointing at the wrong establishment, which no checksum can detect. When absent, the mass splits into SIREN-only, invalid-checksum, and both-missing outcomes.
- **Buyer name.** Either a generic-collision replacement drawn from a type-conditioned pool, or a name variant produced by one of six transformations (abbreviation, suffix removal, punctuation and accent flattening, service qualifier, geographic qualifier, token reorder). Rates are the empirical false-split and false-merge rates, scaled by the notice's severity and the buyer's alias propensity. Name-comparison behaviour of this kind is the classic string-similarity setting of Cohen et al. [3], Jaro [6].
- **CPV.** A schema-and-type conditional missing rate, plus parent-code replacement, division-only truncation, generic-code substitution, and wrong-related-code substitution.
- **Duration.** Presence from the smoothed conditional rate; when present, the visible value is drawn from the empirical distribution of valid raw BOAMP durations by a deterministic low-discrepancy inverse-CDF map keyed on the notice identifier. The latent cycle length is *not* rounded into the observed field — observed duration and true duration are separate quantities, which is the point.
- **Text.** Exact cross-buyer template substitution, near-boilerplate with row-specific tokens, or generic weak procurement wording, at a rate conditional on notice type and buyer activity tier.
- **Linked-notice visibility.** An award's reference back to its call is dropped at the empirical unresolved-award rate.

Missingness terminology. In the taxonomy of Rubin [11] and Carpenter and Smuk [1], none of these mechanisms is missing completely at random. Identifier, CPV and duration presence are missing at random conditional on observed schema, year and notice type; text degradation is additionally conditional on buyer activity tier. The shared quality class Q_ν induces dependence *between* fields, so the joint missingness pattern is not the product of its marginals — which is exactly the property a linkage benchmark must have and an independent-Bernoulli design would lose.

Identity persistence. Observed buyer identity is drawn once per notice family and reused across its members, and — under the scoped block — across the cycles of one hidden need. This is a labelled assumption, not an oversight: without it, sibling notices of one procurement episode drift apart in buyer key and the benchmark becomes unreachable under production-style buyer-key blocking. It is recorded in the mechanism trade-off log as an explicit reachability-versus-conditional-fidelity trade.

11 Detailed generation algorithm

Table 5: Executable order, with the module and the conditioning set for each step. Every stochastic step consumes only ω or only ε, never both.

#	Step	Module	Stream	Conditions on
1	Buyers	`buyers.py`	ω	observation window, scenario Pareto parameters
2	Establishments	`establishments.py`	ω	buyer SIREN, activity tier
3	Needs	`needs.py`	ω	buyer activity weight, empirical CPV division distribution, scoped block
4	Cycles	`cycles.py`	ω	need recurrence propensity and duration profile, gap mixture, window
5	Relations	`relations.py`	—	realised cycle chains only (deterministic)
6	Notices	`notices.py`	ω	cycle dates, empirical by-year eForms adoption share
7	Clean values	`text_generation.py`	ω	buyer, establishment, need vocabulary
8	Corruption	`corruption.py`, `missingness.py`	ε	clean values, schema, year, notice type, activity tier, quality class
9	Export	`pipeline.py`	—	all of the above

Structural validation runs between steps 7 and 8 and raises rather than warns: a world that fails it never reaches corruption. Column-schema validation and the truth-leakage assertion run on the observed table before it is written.

12 Scenario definitions

7 scenarios are configured and 6 have generated artifacts; 1 remains configuration-only and is recorded as such in `registries/scenario_manifest.csv` so a loadable scenario cannot be mistaken for multi-seed evidence. `clean_sanity` is a single-seed smoke test; each of the four bracketing scenarios and the central scenario carries 10 world seeds, for 51 artifacts in total.

Scenarios exist because the recurrence process is unidentified. They are not sensitivity analyses around a believed value; they are a declared envelope of assumptions, and any statement that depends on which scenario is used must say so.

13 Reproducibility and random-seed management

The central world is generated from world seed `20260721` and corruption seed `20260722`, with $B =$ 19,200 buyers. Replicates vary only the world seed; the corruption seed follows it deterministically. Every artifact directory carries a `generation_metadata.json` recording both seeds, the resolved scenario and defaults, SHA-256 hashes of every configuration file and of the calibration corpus, row counts, the Git commit, and — new in this revision — the exact numeric-library versions.

Why the environment is recorded. Replay compares canonical table *content*, not Parquet bytes, so writer-version differences are already tolerated. It does not tolerate a different NumPy: the generator and reduction kernels are bit-stable per build, not across builds, so a different NumPy reproduces every count and identity while differing in the last unit in the last place of float columns — enough to change a SHA-256. This was observed in practice during the audit that produced this revision. The fix is twofold: `requirements-lock.txt` pins the exact environment, and the replay comparison now reports environment drift explicitly so the next such mismatch is diagnosable instead of looking like generator drift. The current artifacts were generated under Python 3.12.3, NumPy 2.4.6, pandas 2.3.3, PyArrow 24.0.0.

Current replay status. Canonical replay of the central world: `PASS` across 9 tables, environment drift none. All-artifact replay: `PASS`, 51/51.

14 Generated datasets and schemas

Table 6: Row counts of the current central world. Sample: scenario `central_provisional`, world `001`, corruption `001`.

Table	Rows	Table	Rows
`latent_buyers`	19,200	`clean_notices`	87,360
`latent_establishments`	20,811	`observed_notices`	87,360
`latent_needs`	44,063	`corruption_log`	291,855
`latent_cycles`	63,291	`notice_family_membership`	87,360
`true_relations`	63,291		

The full column-level dictionary — 83 columns with dtype, null share, distinct count, and visibility flags — is in `registries/data_dictionary.csv`.

15 Hidden ground-truth structure

`true_relations.parquet` holds one row per cycle: `NEXT_CYCLE` with a target and a true gap, or `NO_SUCCESSOR`. A cycle is observed through its call notice, so a cycle-level successor relation becomes a notice-level true match between two call notices. Of the resulting true matches, 1,807 have both notices inside the algorithm's own candidate-generation scope.

That scoping is deliberate and is reported separately rather than folded into recall. A successor whose two notices are not both in scope cannot be found by any method operating on those candidates; charging every probe for the same fixed scoping loss would hide the differences between them, which is the only thing the benchmark exists to reveal.

16 Validation methodology

The framework evaluates 231 metrics grouped into 15 gates, of which 8 are critical. A critical gate fails the release on its own; a non-critical gate can only downgrade the run to `PASS_WITH_WARNINGS`. This asymmetry is intentional: the benchmark is allowed to simplify the real corpus provided the simplification is documented in the fidelity budget, but anything that would silently change a linkage conclusion, leak truth, or copy a real record is critical.

Consequence, and why the headline is not the whole story. A non-critical gate reports `WARNING` even when individual metrics inside it failed. The run therefore reports the metric-level failure count explicitly: 5 failures, 5 of them inside non-critical gates. Reading only `PASS_WITH_WARNINGS` would understate what the benchmark is carrying.

Diagnostics and what each is for.

- **TV and JS** (Eq. 2) for categorical composition: bounded, assumption-free, and interpretable as disagreement mass. Limitation: insensitive to *which* categories disagree, so they are always read alongside the per-cell table.
- **Scaled** W_1 for continuous fields: sensitive to location and spread, expressed in interquartile-range units so different fields are comparable. Limitation: one number cannot distinguish a shifted distribution from a differently-shaped one, which is why the duration comparison is also shown as a quantile plot (Figure 8).
- **Weighted MAE over cells** (Eq. 3) for conditional rates: the correct object when the mechanism, not the average, is what must match. Limitation: it averages, so a large error in one important cell can be masked; the cell-level tables are retained.
- **Pairs completeness, pairs quality, reduction ratio** for blocking: the standard triple [2]. Pairs completeness is recall of true matches into the candidate set; pairs quality is the share of candidates that are true matches; reduction ratio is how much of the quadratic comparison space was avoided.
- **Overlap coefficient** between the score distributions of true matches and hard negatives: the benchmark must be neither trivially separable nor hopeless, so this metric is gated on *both* sides.
- **Cluster bootstrap** resampling buyers for interval estimates, because notices within a buyer are not independent.
- **Nearest-neighbour text overlap** against a real baseline for privacy: the synthetic corpus must not be closer to real text than real text is to itself.

Four things that are not the same. *Statistical significance* is uninformative at this sample size and is not used. *Practical discrepancy* is the effect size against a pre-declared tolerance. *Observable fidelity* is agreement on what a linker can see. *Structural plausibility* is whether the latent mechanism is defensible — which fidelity cannot establish, since many recurrence processes yield similar observed tables. *Suitability for linkage evaluation* is a further step again, and is what the readiness assessment, not the validation status, decides.

Table 7: Validation-framework gate summary for the current run. A critical gate fails the release on its own; a non-critical gate can only downgrade the run to `PASS_WITH_WARNINGS`. Source: `validation_gate_summary.csv`.

Gate	Critical	Pass	Warn	Fail	Status
`internal`	yes	35	0	0	PASS
`specification_recovery`	yes	6	0	0	PASS
`marginals`	no	16	2	0	WARNING
`conditionals`	yes	4	0	0	PASS
`temporal`	yes	8	1	0	WARNING
`candidate_environment`	yes	10	0	0	PASS
`missingness_text_identifier`	no	5	3	2	WARNING
`buyer_activity`	no	10	1	1	WARNING
`missingness_structure`	no	4	0	0	PASS
`names_identifiers`	no	5	1	2	WARNING
`text`	no	9	2	0	WARNING
`privacy`	yes	4	0	0	PASS
`hidden_truth_difficulty`	yes	12	0	0	PASS
`algorithm_utility`	yes	34	0	0	PASS
`robustness`	no	47	6	0	WARNING

17 Synthetic-data results

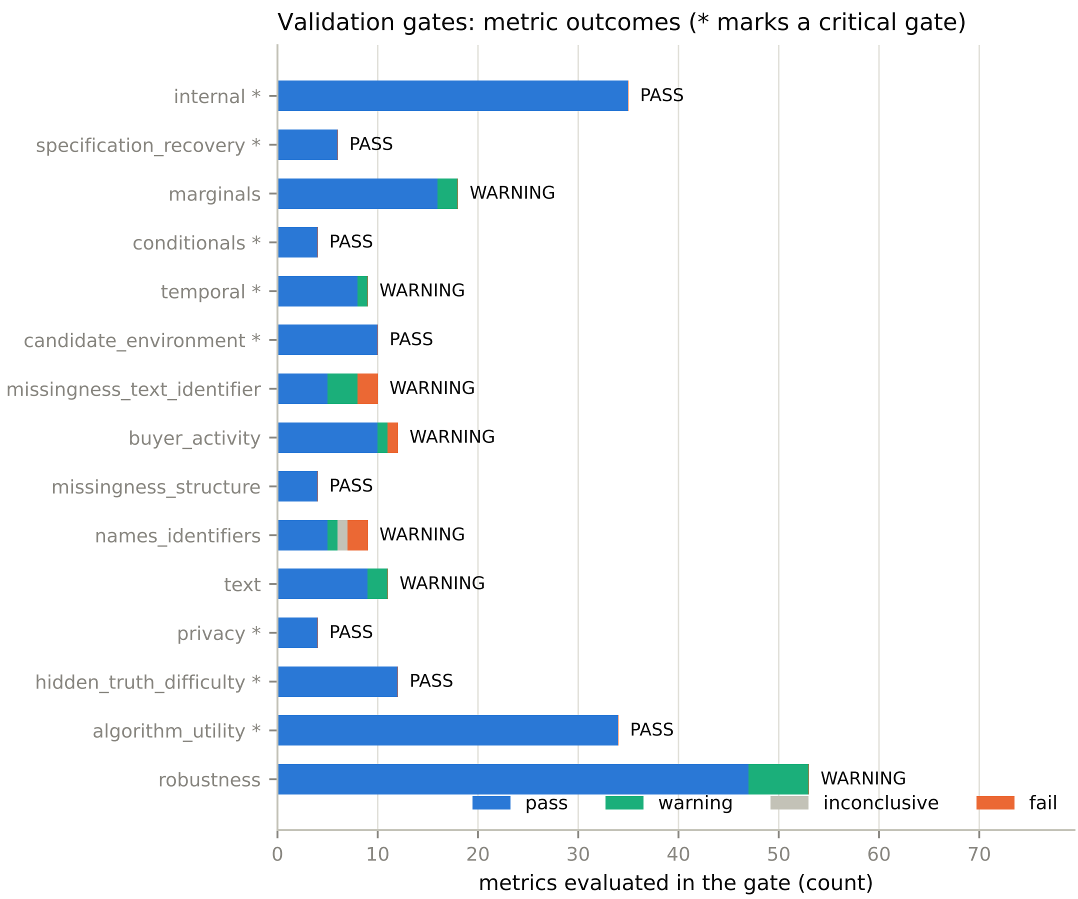

Figure 3: Metric outcomes per gate. Reading this alongside Table 7 is the intended use: the bar shows how much evidence each gate rests on, the label shows the verdict, and a warning on a wide bar is a different situation from a warning on a narrow one.

Of 231 metrics, 209 pass, 16 warn, 5 fail, and 1 are inconclusive. Table 8 lists every non-passing metric with its real and synthetic estimate and the tolerance it was judged against.

Table 8: Every metric that did not pass, with its real and synthetic estimates and the tolerance it was judged against. Long names are truncated for width; the untruncated table is `discrepancy_register.csv`. A blank real estimate marks a quantity that exists only inside the synthetic truth and therefore has no real counterpart.

Gate	Property	Metric	Real	Synth.	Tol.	Status
buyer activity	relative notices per buyer	q99 abs diff	16.206	13.864	1.0	Fail
missingness text iden...	siret missing	rate abs diff pp	0.728	0.687	2.0	Fail
missingness text iden...	siret present	presence rate abs dif...	0.272	0.313	2.0	Fail
names identifiers	name token jaccard (silver siren ...	q10 abs diff	0.000	0.333	0.15	Fail
names identifiers	name token jaccard (silver siren ...	q50 abs diff	0.250	0.667	0.15	Fail
names identifiers	siren invalid among present	abs diff pp		0.000	2.0	Inconclusive
buyer activity	activity share (top1pct)	abs diff pp	0.383	0.331	10.0	Warning
marginals	text length	W1 scaled			0.1	Warning
marginals	text length	q75 relative error	133.000	118.000	0.1	Warning
missingness text iden...	siren missing	rate abs diff pp	0.728	0.694	2.0	Warning
missingness text iden...	cpv missing	rate abs diff pp	0.171	0.150	2.0	Warning
missingness text iden...	siren present	presence rate abs dif...	0.272	0.306	2.0	Warning
names identifiers	name jaro winkler (silver siren g...	q50 abs diff	0.695	0.885	0.15	Warning
robustness	blocking pairs completeness (cros...	coefficient of variat...		0.301	0.25	Warning
robustness	probe headroom (cross scenario)	coefficient of variat...		0.570	0.25	Warning
robustness	probe ranking (ranking stability:...	min kendall tau		0.333	0.8	Warning
robustness	probe ranking (ranking stability:...	min kendall tau		0.333	0.8	Warning
robustness	probe ranking (ranking stability:...	min kendall tau		0.333	0.8	Warning
robustness	probe ranking (ranking stability)	min kendall tau		0.333	0.8	Warning
temporal	followup runway (60m)	abs diff pp	0.550	0.440	5.0	Warning
text	unigram distribution	JS			0.3	Warning
text	bigram distribution	JS			0.4	Warning

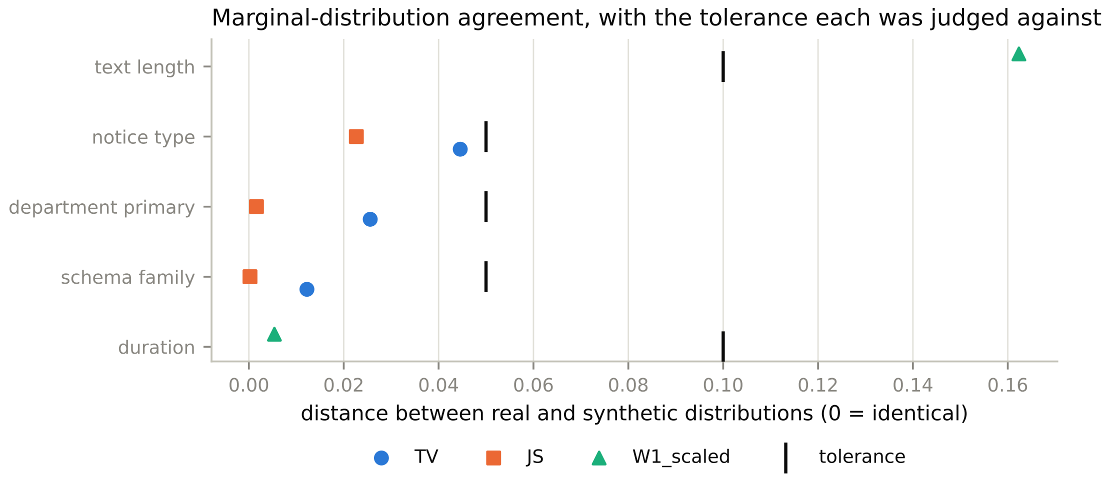

Figure 4: Agreement of each observable categorical marginal. Both distances are bounded in $[0, 1]$; values near zero mean the two corpora would be hard to tell apart on that variable alone.

18 Comparison with the original BOAMP data

All comparisons below apply the *same* definitions, filters, grouping rules, and units to both corpora. Where a quantity is computable only inside the synthetic truth, it is labelled as such and no real counterpart is shown.

18.1 Candidate environment (directly comparable)

Table 9: Per-source candidate-set size under the production Layer-1 blocker, real corpus versus two generator versions. Unit: candidates per eligible source notice. Source: `candidate_environment_summary_metrics.csv`.

Scope	n sources	Zero-cand.	Mean	Median	$q_{.75}$	$q_{.90}$	$q_{.95}$	Cap hit
Real BOAMP, Layer 1 scope	3,380	40.7%	3.21	1	4.0	9.1	14.0	0.09%
Synthetic v0.2	354	90.4%	0.15	0	0.0	0.0	1.0	0.00%
Synthetic v0.3 (current)	3,920	42.4%	2.28	1	3.0	7.0	10.0	0.00%

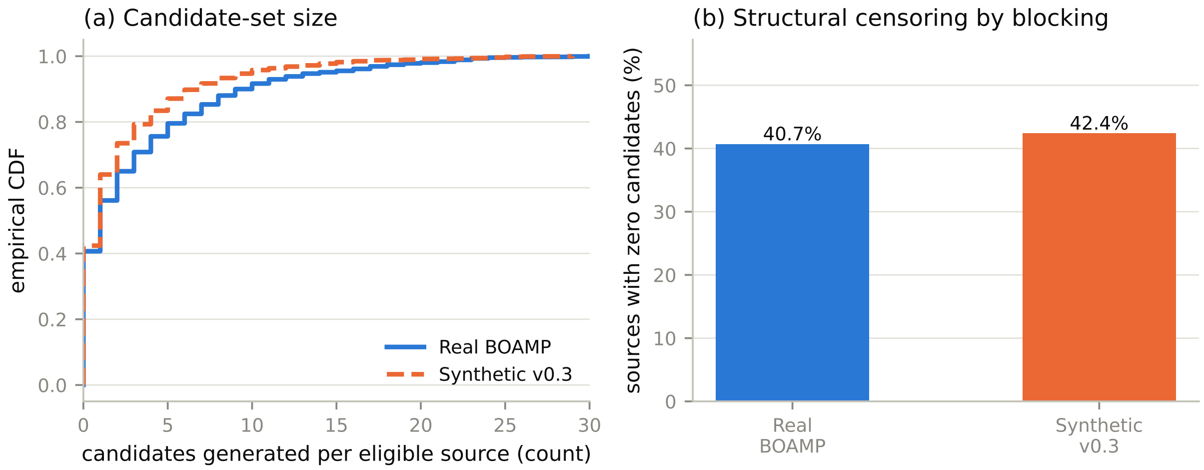

Figure 5: Candidate-set size and structural censoring. Panel (a) is the empirical CDF of candidates per source; panel (b) is the share of sources the blocker leaves with nothing to score. Panel (b) is the more important of the two: a source with no candidate is censored by construction, and no scoring improvement can recover it.

The v0.3 environment clears every predefined gate (7/7 dimensions): source-count ratio 1.123 against the scaled real target, zero-candidate absolute difference 1.69 percentage points, and upper-quantile absolute differences of 1.0, 2.1 and 4.0 candidates at the 75th, 90th and 95th percentiles. Neither corpus reaches the per-source candidate cap of 30 at a material rate. The contrast with v0.2 — 90.4% zero-candidate — is the single largest improvement in this version.

18.2 Buyer-level structure (directly comparable)

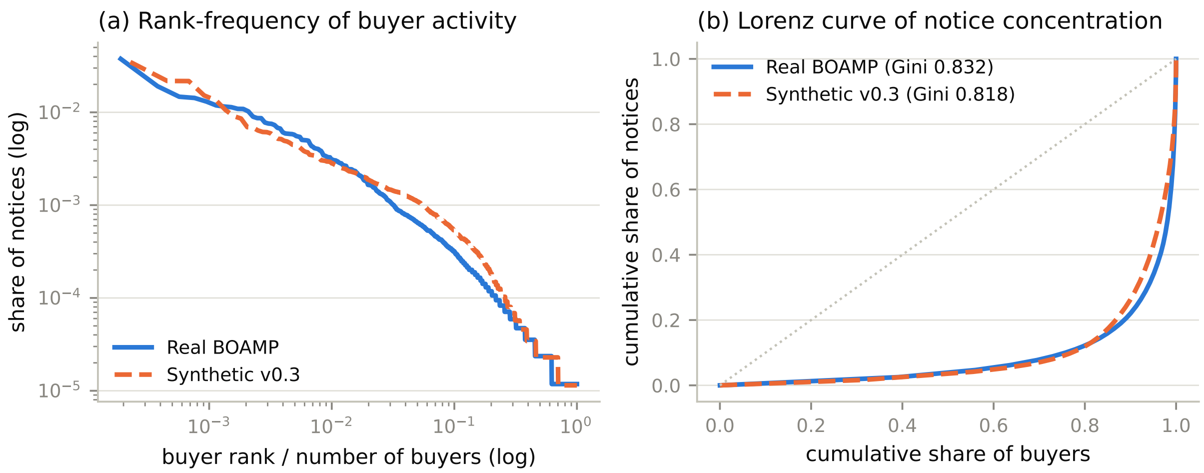

Figure 6: Notice concentration across buyers. Note the deliberately different identity definitions: the real curve uses the production buyer key, the synthetic curve uses the observed buyer name — the identity a linker actually has. Using latent buyer identity for the synthetic side would compare quantities with different denominators and flatter the generator.

18.3 Conditional observation mechanisms (directly comparable)

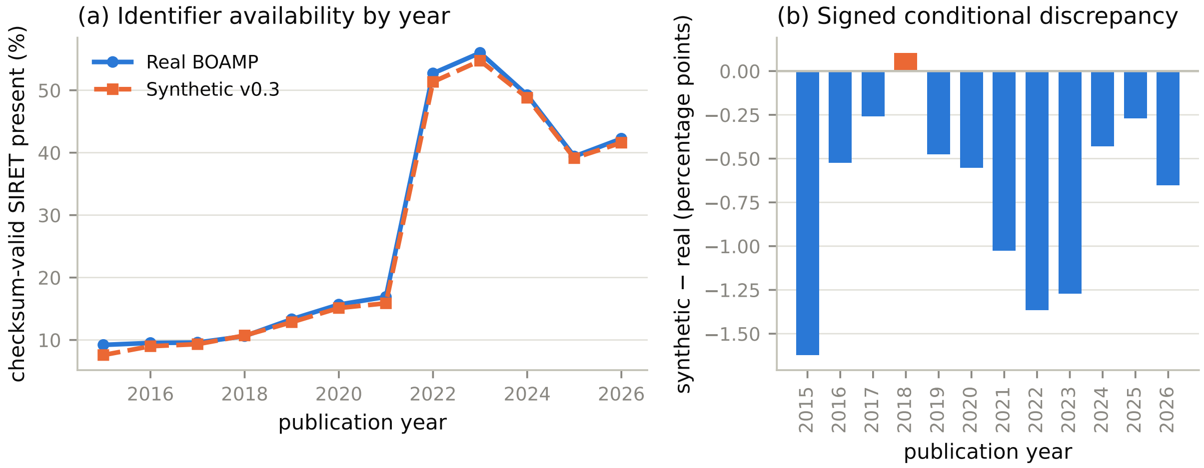

Figure 7: Checksum-valid SIRET presence by publication year. Panel (b) is the signed discrepancy in percentage points; the sign matters, because a generator that is uniformly optimistic about identifier availability makes linkage systematically easier than reality.

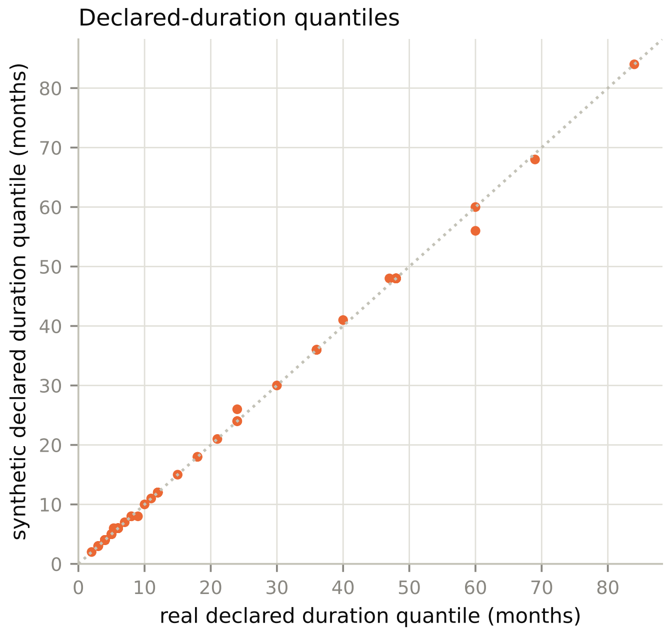

Figure 8: Declared-duration quantiles, real against synthetic. This compares the *observed* declared duration only. The latent contract length is generated by a separate mechanism and has no real counterpart — conflating the two would be a category error.

18.4 Quantities available only inside the synthetic truth

Blocking recall, recovery by corruption type, cluster-level accuracy, and every probe-linker score in Section 19 exist only because truth was generated. There is no real-BOAMP number to place beside them, and any table that appeared to do so would be comparing a measurement with an assumption.

19 Linkage-difficulty analysis

Blocking dominates. Of 1,807 in-scope true successor pairs, the production window recovers 0.392 and the widened 36-month window recovers 0.956. The gap is a property of that window against these gaps, not a generator defect: the gap distribution is deliberately centred outside the production window (Equation 11), because a benchmark whose true successors all fell inside the window would report a blocking recall that says nothing about renewal timing.

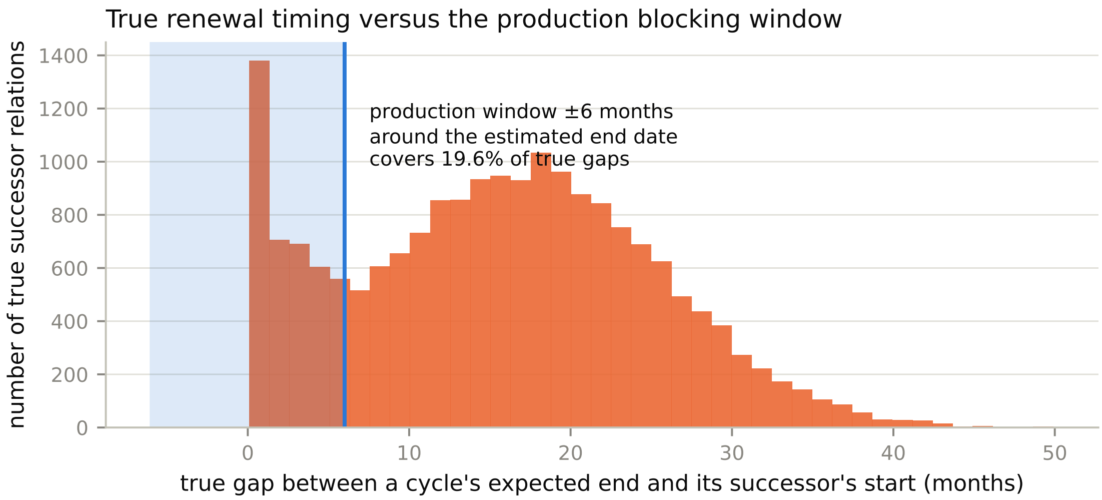

Figure 9: True renewal gaps against the production blocking window. Everything outside the shaded band is unreachable by the production blocker regardless of how good the comparison model is.

Recall decomposes exactly. End-to-end recall factorises as

$$R_{\text{end-to-end}} = R_{\text{blocking}} \times R_{\text{scoring}|\text{reachable}}, \tag{13}$$

and the framework asserts this identity numerically rather than assuming it, so a probe's weakness is attributable to the stage that caused it. Pairs quality in the production window is 0.079 and the reduction ratio is 0.999: the blocker discards almost all of the quadratic comparison space, at the cost of also discarding most true matches.

The task is discriminative. The overlap coefficient between the score distributions of true matches and hard negatives is 0.437 — inside the gated band on both sides, so the task is neither trivially separable nor hopeless. Mean reciprocal rank of the true match among a source's candidates is 0.904, and the median top-1-to-top-2 margin is 0.165: when the true match is present it usually ranks first, but by a margin small enough that the threshold choice matters.

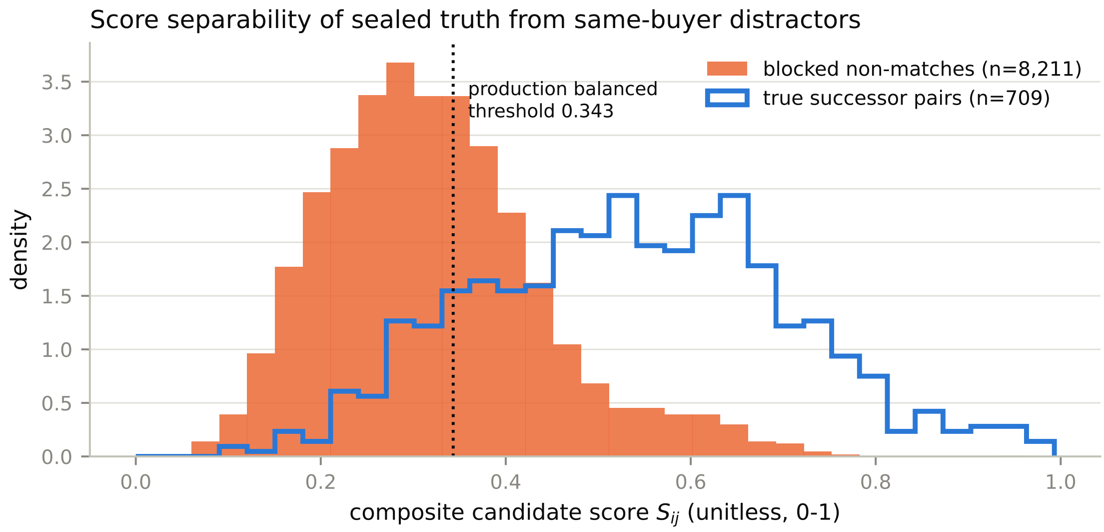

Figure 10: Composite production score for blocked pairs, split by sealed truth. The overlap of the two densities is what a comparison model must resolve; the vertical line marks the production acceptance threshold, shown for reference and not used to define truth.

Table 10: Frozen probe linkers scored against sealed synthetic truth on the central world. These are benchmark-discrimination diagnostics on generated truth, not estimates of real BOAMP accuracy. Source: `probe_linker_results.csv`.

Probe	Predicted pairs	Pair prec.	Pair rec.	Pair F_1	B-cubed F_1
`deterministic_top1`	1,024	0.438	0.248	0.316	0.730
`score_threshold_all_ranks`	768	0.436	0.185	0.260	0.737
`fellegi_sunter`	1,854	0.188	0.193	0.191	0.694

Three frozen probe linkers are run — a deterministic top-1 rule, a score-threshold rule over all ranks, and a Fellegi–Sunter weight model [4]. Their parameters were chosen once and never tuned against these results, and none of them is the production threshold. Best pair F_1 is 0.316 (`deterministic_top1`), comfortably inside the gated band: a benchmark on which every probe scored near 1 or near 0 would not discriminate between methods.

Note the divergence between pair-level and cluster-level scores: the score-threshold probe attains pair $F_1 = 0.260$ but B-cubed $F_1 = 0.737$. Most notices are singletons, so cluster-level metrics are dominated by correctly leaving them alone. Both are reported because they answer different questions, and quoting only the flattering one would mislead.

20 Sensitivity and robustness analysis

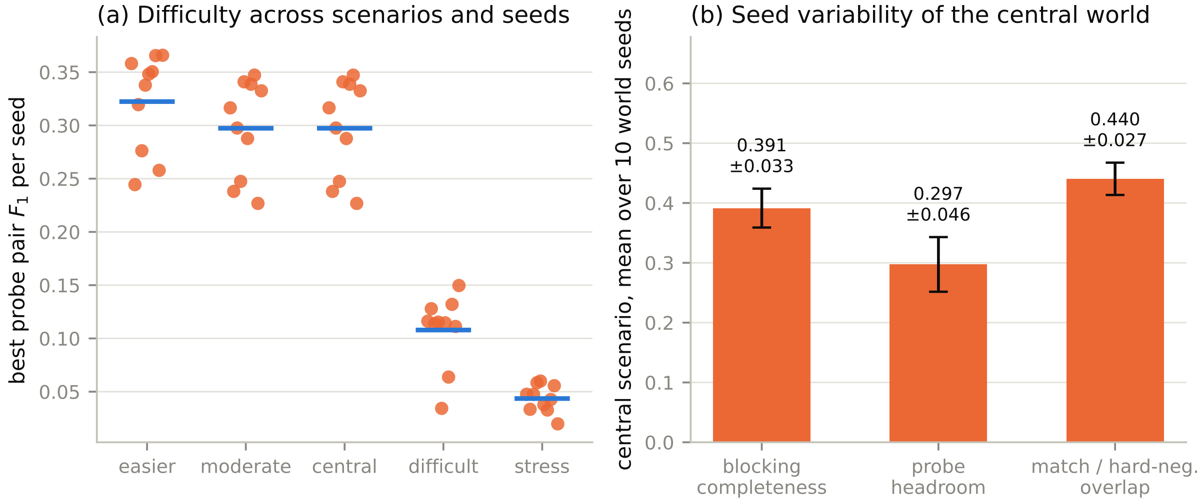

Figure 11: Seed-to-seed variability. Panel (a): best probe pair-F_1 per generated world, one point per seed, bar at the scenario mean. Panel (b): mean and standard deviation of the difficulty metrics over the 10 central seeds. This figure is the reason the readiness assessment blocks controlled comparison.

Over 10 central world seeds, production blocking completeness has mean 0.391, standard deviation 0.033, worst case 0.341, and coefficient of variation 0.083. Probe headroom has mean 0.297, standard deviation 0.046, and coefficient of variation 0.154. Both coefficients exceed the declared tolerance.

150 replicate-level probe results across 51 artifacts feed the ranking-stability summary. Probe rank order is not stable within a scenario, let alone across scenarios. That is the decisive finding of this section: a benchmark whose ranking flips with the seed cannot rank algorithms, however internally correct it is.

21 Known limitations

1. **Recurrence is unidentified.** Prevalence, gap distribution, text drift, and the specific CPV corruption mode are assumptions. Several distinct latent processes produce similar observed tables, so these parameters are partially identified at best.
2. **Seed instability.** Difficulty metrics and probe rankings vary materially across seeds; see the previous section.
3. **In-sample fidelity.** No real-BOAMP partition was held out from generator design, so the fidelity results are not out-of-sample [12].
4. **Residual metric failures.** 5 metric-level failures remain, all in non-critical gates; see Table 8.
5. **Scope.** One region, one sector, two notice types. The residual `OTHER` notice type and the modification/cancellation lifecycle are not modelled.
6. **Identity persistence.** Observed buyer identity is held constant within a notice family and, under the scoped block, across a need's cycles. This is an explicit reachability-versus-fidelity trade, not a neutral choice.
7. **Placeholder taxonomy.** No manually validated technological taxonomy exists in the project, so needs use broad CPV-division segments.

8. **Untested attack surfaces.** Membership-inference attacks [13], neural-embedding semantic distance, and variance-based global sensitivity analysis are out of scope and are nowhere reported as passed.

22 Appropriate and inappropriate uses

Table 11: Five-level readiness assessment. This, not the validation headline, governs what the benchmark may currently be used for. Source: `readiness_decisions.json`.

Readiness level	Status	Why blocked
`PIPELINE_TECHNICALLY_VALID`	PASS	none
`READY_FOR_PRELIMINARY_MODELING`	PASS_WITH_LIMITATIONS	none
`READY_FOR_CONTROLLED_ALGORITHM_COMPARISON`	FAIL	observable hard-fidelity failures remain in non-critical current gates
`READY_FOR_FINAL_ALGORITHM_RANKING`	FAIL	observable hard-fidelity failures remain in non-critical current gates, headline difficulty varies materially across...
`READY_FOR_VALIDATED_SYNTHETIC_BENCHMARK_RELEASE`	FAIL	observable hard-fidelity failures remain in non-critical current gates, headline difficulty varies materially across...

Supported now. Debugging a linkage implementation end to end; integration testing against a realistic schema; measuring how a method's blocking recall responds to a known change in identifier availability; demonstrating that a method degrades in the expected direction between the `easier` and `stress` scenarios.

Not supported now. Ranking two linkage methods; selecting an acceptance threshold; quoting any precision or recall number as an estimate of real-BOAMP performance; describing the benchmark as validated or released; citing synthetic recurrence prevalence as evidence about real renewal behaviour.

Never supported. Substituting for the real BOAMP ground truth. The truth in this benchmark is generated. It is a correct answer to a constructed question, and its usefulness rests entirely on the conditional claim in Section 2 — which the readiness assessment currently rates as not yet established for comparison purposes.

23 Reproducibility instructions

1. **Environment.** `python -m venv .venv && .venv/bin/pip install -r requirements-lock.txt`. The lock, not `requirements.txt`, is what makes replay bit-exact.
2. **Retrieval** (only if `data/raw` is empty): `scripts/download_boamp.py`, `scripts/download_enrichment.py`.
3. **Real pipeline and calibration:** execute `notebooks/01_data_engineering_pipeline.ipynb`, then `notebooks/04_synthetic_benchmark_calibration.ipynb`. The second writes every table in `reports/tables/synthetic_calibration/` that the generator consumes, so it must precede generation.
4. **Generation:** `scripts/generate_synthetic_benchmark_replicates.py` for the bracketing scenarios and the central seeds, then `scripts/generate_synthetic_benchmark_v0_3_temporal_candidate_revision.py`, which selects the scoped parameters and writes the central world with its selection provenance.
5. **Validation and readiness:** `validate_synthetic_benchmark.py`, `replay_synthetic_benchmark.py`, `replay_synthetic_benchmark_replicates.py`, `build_benchmark_registries.py`, `build_mechanism_tradeoff_log.py`, `assess_synthetic_benchmark_readiness.py`.
6. **This report:** `build_report_values.py`, then `build_report_figures.py`, then `latexmk -pdf` on `reports/synthetic_benchmark_technical_report.tex`.
7. **Consistency gate:** `check_report_consistency.py` re-derives every reported value from the artifacts, verifies that no macro is undefined and no figure missing, and fails if the document is stale.
8. **Tests:** `pytest -q` and `pytest -q -m slow`.

The ordering constraint that is easy to get wrong: the replicate generator must run *before* the central sweep script, because both write world 001 of the central scenario and only the sweep script records the parameter-selection provenance in its metadata.

24 Conclusion

The benchmark is internally correct and reproducible. Its latent world is generated in a declared order from two independent seeds; its truth is a deterministic function of realised cycle chains; no truth value, alias, or hash reaches the observed table; and every one of the 51 generated artifacts replays to identical canonical hashes under the pinned environment.

Its observable realism is a measured, bounded claim rather than an assumption. The candidate environment — the property that decides whether linkage is possible at all — now matches the real corpus within the declared tolerances, and the conditional observation mechanisms for identifiers, CPV, duration and text reproduce the real corpus's structure rather than only its averages. Where they do not, the residual is listed in Table 8 rather than smoothed away.

What remains unresolved is not fidelity but stability. Difficulty metrics and probe rankings move materially between world seeds, and until they do not, the benchmark cannot support controlled algorithm comparison or algorithm ranking. That is the honest current position, and it is what the five-level readiness assessment records.

Finally, the framing that should survive every use of this artifact: the recurrence relation here is *generated*, not discovered. It is not a reconstruction of what French public buyers actually do, and it is not a replacement for the BOAMP ground truth that does not exist. It is a controlled instrument for measuring how a linkage method behaves when the answer is known, under a set of assumptions this report has tried to make impossible to overlook.

A Technical appendices

A.1 Frozen production algorithm parameters

These describe the linkage pipeline the benchmark is measured against. They are *algorithm* parameters: they may be reported for operational comparison and never define synthetic truth.

Table 12: Production linkage settings. Source: `config/pipeline.yaml`.

Parameter	Value
Score weights $w_{\text{text}}, w_{\text{cpv}}, w_{\text{time}}, w_{\text{buyer}}$	0.35, 0.30, 0.25, 0.10
Acceptance threshold τ (balanced)	0.343167
Sensitivity thresholds (broad / strict)	0.278387 / 0.442070
Temporal half-window W	6 months
Candidate cap per source	30
Days per month	30.44

The composite score is $S_{ij} = w_{\text{text}}s_{\text{text}} + w_{\text{cpv}}s_{\text{cpv}} + w_{\text{time}}s_{\text{time}} + w_{\text{buyer}}s_{\text{buyer}}$. Weights are fixed a priori, not fitted. The thresholds are percentiles of the Layer-1 rank-1 score distribution, frozen so that layer comparisons are not confounded by threshold re-derivation. Neither is an independently validated cutoff.

A.2 Where each artifact comes from

Table 13: Dependency map for this report. Every row is regenerated by the command in the last column.

Artifact	Depends on	Produced by
`data/interim/boamp_common_prepared.csv`	raw monthly JSON	notebook 01
`data/processed/boamp_only/*`	prepared corpus, `pipeline.yaml`	notebook 01
`reports/tables/synthetic_calibration/*`	prepared corpus, Layer-1 outputs	notebook 04
`config/synthetic/calibration_parameters_v0_1.yaml`	calibration tables (manually frozen)	—
`data/processed/synthetic_benchmark/*`	calibration tables, scenario files, seeds	generator scripts
`validation_framework/*`	generated artifacts, prepared corpus	`validate_synthetic_benchmark.py`
`registries/*`	generated artifacts, scenario files	`build_benchmark_registries.py`
`readiness/*`	validation manifest, replicate inventory	`assess_synthetic_benchmark_readiness.py`
`generated/synthetic_benchmark/report_values.tex`	all of the above	`build_report_values.py`
`figures/synthetic_benchmark/report/*`	generated artifacts, validation metrics	`build_report_figures.py`
this PDF	the two rows above	`latexmk`

A.3 Provenance of this document's numbers

No statistic in this report was typed by hand. Each is a macro defined in `reports/generated/synthetic_benchmark/report_values.tex`, generated by `scripts/build_report_values.py` from the artifacts above, with the same values also written to `report_values.json`. `scripts/check_report_consistency.py` re-derives them and fails if the document and the artifacts have diverged — the mechanism that keeps this report from going stale the way its predecessors did.

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